Handling Data 2 — Frequency tables · Discrete and continuous data

Calculator allowed. Squared paper for the frequency polygons. Marks in brackets. ★ = harder question.

1 Dani records the number of goals (x) she scores per game over a season of 20 games. [5]

x	0	1	2	3	4	5	6
Frequency	1	4	6	5	2	1	1

Last season her mean was 2.1 goals per game. Show that she has improved this season, and write down this season's mode and median.

> $\sum fx = 0 + 4 + 12 + 15 + 8 + 5 + 6 = 50$; $\sum f = 20$. Mean $= \dfrac{50}{20} = 2.5 > 2.1$, so she has improved. Mode $= 2$ (frequency 6). Median: 20 values → between the 10th and 11th; running totals 1, 5, 11 — both are 2, so median $= 2$.

2 The times (t minutes) taken by 20 students to finish a puzzle were: [10]

3, 16, 5, 19, 6, 7, 10, 8, 13, 12, 9, 7, 10, 11, 12, 16, 15, 13, 8, 5

a Make a frequency table with classes $0 \le t < 5$, $5 \le t < 10$, $10 \le t < 15$, $15 \le t \le 20$.

b Draw a frequency polygon.

c Use the table to estimate the mean time, and state the modal class.

d Calculate the exact mean from the raw data and explain why it differs from your estimate.

> **a** Frequencies: $0 \le t < 5$: 1 · $5 \le t < 10$: 8 · $10 \le t < 15$: 7 · $15 \le t \le 20$: 4 (total 20 ✓).
>
> **b** Plot frequency against the class mid-points $2.5,\ 7.5,\ 12.5,\ 17.5$ and join with straight lines.
>
> **c** $\sum fx = 1(2.5) + 8(7.5) + 7(12.5) + 4(17.5) = 2.5 + 60 + 87.5 + 70 = 220$; estimated mean $= \dfrac{220}{20} = 11.0$ min. Modal class $5 \le t < 10$.
>
> **d** Raw total $= 205$, exact mean $= 10.25$ min. The estimate uses the mid-point of each class instead of the real values — here most values sit below their class mid-points, so the estimate is slightly too high.

3 A survey of 100 teenagers asked how many hours (t) they slept on New Year's Day. [10]

Hours t	$0 \le t < 4$	$4 \le t < 6$	$6 \le t < 8$	$8 \le t \le 12$
Frequency	p	24	$3p$	16

a Find p and complete the table.

b Draw a frequency polygon.

c Estimate the mean and state the modal class.

d Estimate the percentage of teenagers who slept at least 7 hours.

a $p + 24 + 3p + 16 = 100 \Rightarrow 4p = 60 \Rightarrow p = 15$. Frequencies 15, 24, 45, 16.
b Mid-points 2, 5, 7, 10 against frequency.
c $\sum fx = 15(2) + 24(5) + 45(7) + 16(10) = 30 + 120 + 315 + 160 = 625$; mean $\approx \dfrac{625}{100} = 6.25$ h. Modal class $6 \le t < 8$.
d Assume the $6 \le t < 8$ class is spread evenly: half of 45 $= 22.5$ slept 7–8 h, plus all 16 of the last class: $38.5 \approx 39\%$.

4 The mean of the numbers 4, 7, x, 9 and 11 is 8. Find x, and then find the median. [3]

Total $= 8 \times 5 = 40$, so $x = 40 - 31 = 9$. In order: 4, 7, 9, 9, 11 → median 9.

Total: 28 marks

Check: mean from a table $= \dfrac{\sum fx}{\sum f}$ (divide by the total frequency, not the number of rows) · grouped data → mid-points, and the mean is an *estimate* · modal class = highest frequency · frequency polygon: points at mid-points, joined by straight lines.